Randomized Clinical Trial to Compare Negative-Pressure Wound Therapy Approaches with Low and High Pressure, Silicone-Coated Dressing, and Polyurethane Foam Dressing

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Background: This study was designed to compare two approaches to negative-pressure wound therapy: 125-mmHg pressure with a polyurethane foam dressing and 75-mmHg pressure with a silicone-coated dressing.
Methods: Forty patients with diabetic foot wounds, after incision and drainage or amputation for infection, were assigned randomly to negative-pressure wound therapy with 75-mmHg continuous pressure with a silicone-covered dressing (75-mmHg group) or 125-mmHg with a polyurethane foam dressing (125-mmHg group) for up to 4 weeks or until surgical closure.
Results: There was no difference in the proportion of wounds that were closed surgically (75-mmHg group, 50 percent; 125-mmHg group, 60 percent), wounds that demonstrated 50 percent wound area reduction (75-mmHg group, 65 percent; 125-mmHg group, 80 percent), or wounds that demonstrated 50 percent wound volume reduction after 4 weeks of therapy (75-mmHg group, 95 percent; 125-mmHg group, 90 percent).
Conclusion: The authors' results suggest that there was no difference in outcomes in wounds treated with low pressure (75 mmHg) with a silicone-coated interface and high pressure (125 mmHg) with a polyurethane foam interface. (*Plast. Reconstr. Surg.* 133: 722, 2014.)
CLINICAL QUESTION/LEVEL OF EVIDENCE: Therapeutic, II.

Negative-pressure wound therapy has dramatically changed the treatment of complex wounds. Randomized controlled trials have shown that negative-pressure wound therapy increases the proportion of wounds that heal and the median time to heal.[1,2] Recently, there have been several new negative-pressure wound products with different wound interfaces and recommendations for different pressure levels. The most commonly used pressure is based on animal studies[3] that showed that local tissue perfusion was optimized at 125 mmHg. More recent animal studies have reported different findings. For instance, Borgquist et al. used a similar swine model with a gauze dressing and reported that pressures up to 80 mmHg increased local perfusion and higher pressures reduced perfusion.[4–6] In contrast, Timmers et al. measured perfusion on the forearm of healthy adults with negative-pressure wound therapy and found an increase in perfusion up to 300 mmHg, with no deleterious effects up to

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500 mmHg.[7] If studies that model cutaneous perfusion as a surrogate for wound healing report contradictory findings, what pressures should be considered to treat diabetic foot wounds?

The same dilemma is true when selecting negative-pressure wound therapy dressings. Polyurethane foam dressings have been the mainstay of this therapy. Borgquist et al. demonstrated essentially identical results when foam and gauze wound interfaces were used with negative-pressure wound therapy in a swine model.[6,8,9] Currently, there are no clinical studies that compare different pressures or different dressing with negative-pressure wound therapy. The objective of this study was to compare two approaches to negative-pressure wound therapy: 125 mmHg of continuous pressure with a polyurethane foam dressing (V.A.C. with GranuFoam dressing; Kinetic Concepts, Inc., San Antonio, Texas) and 75 mmHg of continuous pressure with a silicone-coated dressing (Engenex with Bio-Dome Technology; ConvaTec, Skillman, N.J.).

PATIENTS AND METHODS

Forty patients with diabetic foot wounds after incision and drainage or amputation for infection were randomized from a computer-generated list to receive negative-pressure wound therapy with 75 mmHg of pressure with a silicone-covered dressing (75-mmHg group) or 125 mmHg of pressure with a polyurethane foam dressing (125-mmHg group) for up to 4 weeks. Both negative-pressure wound therapy devices were changed three times per week. The primary outcomes were 50 and 100 percent wound area reduction.

We included subjects with diabetes aged 21 to 90 years, surgical lower extremity wounds, and ankle-brachial indices greater than 0.70. The two treatment groups demonstrated no statistical difference in age, sex, the presence or duration of diabetes, body mass index, starting wound area or volume, or posterior tibialis or dorsalis pedis artery ankle-brachial index. Digital photographs and wound measurements were obtained once per week. Photographs were scaled and analyzed by ImageJ software (National Institutes of Health, Bethesda, Md.) to calculate wound surface area. Depth was measured at each time point, and volume was calculated using the formula for the volume of a sphere. Medical history included the duration of diabetes, medications (i.e., insulin, oral, combination therapy, and diet), previous history of foot ulcers, or amputation. Sensory neuropathy evaluation consisted of vibratory perception threshold testing and 10-g Semmes-Weinstein monofilaments.[10,11]

Statistical Analysis

The chi-square test was used to compare the proportion of wounds with 50 and 100 percent wound area and volume reduction after 4 weeks of therapy. Time course measurements were analyzed by two-way analysis of variance.

RESULTS

The two treatment groups demonstrated no statistical difference in age, sex, the presence or duration of diabetes, body mass index, starting wound area or volume, or posterior tibialis or dorsalis pedis artery ankle-brachial index (Table 1). The wound area and volume were larger in the 125-mmHg group (75-mmHg group, 20.10 ± 14.33 cm^2; 125-mmHg group, 34.61 ± 32.92 cm^2), but the difference did not attain significance ($p = 0.08$). There was no difference in the proportion of wounds that were surgically closed (75-mmHg group, 50 percent; 125-mmHg group, 60 percent), wounds that demonstrated 50 percent wound area reduction (75-mmHg group, 65 percent; 125-mmHg group, 80 percent), or wounds that demonstrated 50 percent wound volume reduction after 4 weeks of therapy (75-mmHg group, 95 percent; 125-mmHg group, 90 percent). Figure 1 demonstrates that there was no difference in the change in wound volume or area between the two treatment groups for any of the time points. There was one study-related serious adverse event in each treatment group related to infection. There were two non–study-related adverse events in the 75-mmHg group. None of the study subjects voluntarily withdrew from the study.

DISCUSSION

This is the first randomized clinical trial that compares different negative-pressure wound therapy devices and wound interfaces to treat complex diabetic foot wounds. Especially in the early stages of treatment, the most dramatic effect of negative-pressure wound therapy is the reduction in wound depth (volume) and the ability to cover deep structures such as fascia, muscle, tendons, and bone. This is the first randomized controlled trial that evaluates both wound area reduction and wound volume reduction with negative-pressure wound therapy. Our results suggest that there was no difference in wound healing or adverse events among patients treated with low pressure (75 mmHg) with a silicone-coated interface and high-pressure (125 mmHg) with a foam interface.

Table 1. Summary of Demographics, Patient Characteristics, and Clinical Outcomes

Characteristics	75-mmHg NPWT Group	125-mmHg NPWT Group
Demographics		
No.	20	20
Age, yr	50.5 ± 7.4	51.3 ± 12.7
Male sex, %	65	85
BMI, kg/m^2	33.9 ± 10.1	33.7 ± 10.6
Education, yr	11.47 ± 3.75	11.63 ± 4.02
Race, %		
Caucasian	10	25
African American	45	35
Hispanic	40	40
Asian	5	0
Smoking, %		
Nonsmoker	50	40
Previous smoker	30	50
Current smoker	20	10
Marital status, %		
Single	60	45
Married	35	40
Divorced/widowed	20	35
Clinical characteristics		
Type 2 DM, %	90	100
Duration of diabetes, yr	14.8 ± 8.7	11.7 ± 9.4
History of amputation, %	65	65
History of ulcer, %	100	100
Wound area at baseline, cm^2	20.1 ± 14.3	34.6 ± 32.9
Wound volume at baseline, cm^3	35.1 ± 33.0	65.3 ± 69.9
Medication, %		
Oral	20	35
Insulin	50	40
Both	30	25
Posterior tibial ABI	1.01 ± 0.36	1.07 ± 0.18
Dorsalis pedis ABI	1.10 ± 0.30	1.08 ± 0.22
Vibration perception threshold, V	66.0 ± 26.2	68.2 ± 23.7
10-g monofilament, no. of sites missed	7.6 ± 2.8	6.6 ± 4.0
Outcome		
100% wound area-volume reduction, %	50	60
50% wound area reduction, %	65	80
50% wound volume reduction, %	95	90
Adverse events, %		
Study-related	10	5
Non–study-related	5	5
Average days off NPWT	1.9 ± 2.5	1.0 ± 1.6

NPWT, negative-pressure wound therapy; BMI, body mass index; DM, diabetes mellitus; ABI, ankle-brachial index.

Several previous studies have advocated using 4-week wound area reduction rates of 50 or 60 percent[12–14] as a surrogate marker for complete wound closure. The majority of wounds in both groups demonstrated a 50 percent wound area reduction by 4 weeks (75-mmHg group, 65 percent; 125-mmHg group, 80 percent), suggesting that a high rate of complete healing would be expected in 12 weeks. This study differs dramatically from previous negative-pressure wound therapy randomized controlled trials based on the proportion of wounds that were surgically closed. Armstrong and Lavery reported that a small proportion of wounds treated with negative-pressure wound therapy were surgically closed during a 16-week evaluation period (15.6 percent),[1] compared with a majority of wounds that were surgically closed (50 to 60 percent) in this study in 4 weeks. Here, negative-pressure wound therapy was not used for complete wound closure and epithelialization as described in Armstrong and Lavery's negative-pressure wound therapy study. Negative-pressure wound therapy was used to reduce wound depth, cover vital structures, and facilitate drainage so that the wound could be surgically closed when possible. The high rate of surgical closure represents an evolution in the use of negative-pressure wound therapy. Negative-pressure wound therapy is rarely used for a 16-week course of therapy as described in the study by Armstrong and Lavery in 2005.

There are several important limitations to the study. This was a small pilot study and probably underpowered. A few large wounds in one group, in a study with only 40 subjects, may have skewed the severity of the wounds and the clinical

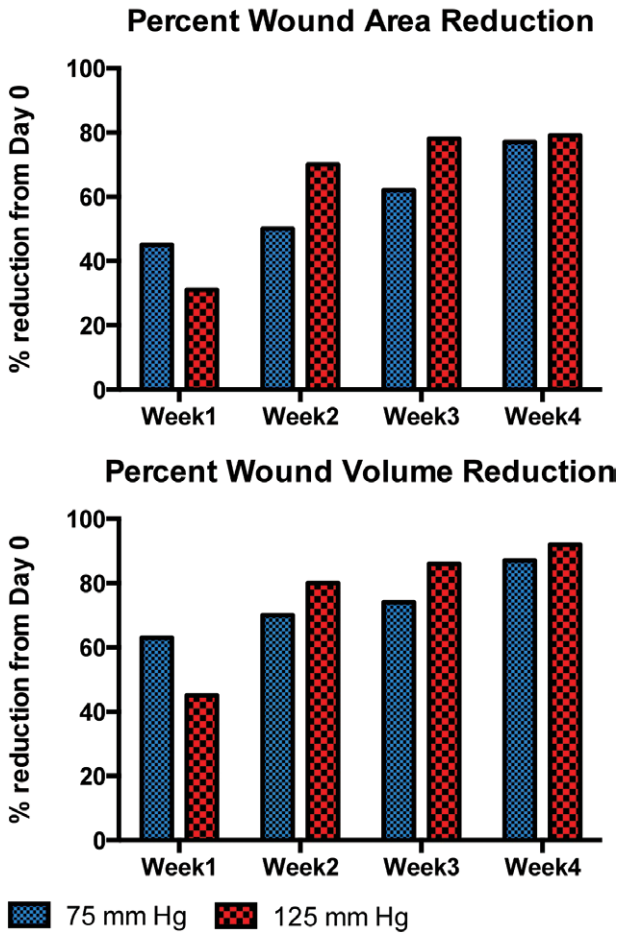

Fig. 1. Percentage wound area and volume reduction in 75-mmHg and 125-mmHg treatment groups. There was no significant difference in wound area or wound volume reduction at any of the time points between the two treatment groups.

outcomes. Even though the difference in wound size was not statistically significant, for most surgeons, the trend is concerning. In addition, two different pressure levels and two different wound dressings were used. This reflects current, "real-world" applications of negative-pressure wound therapy. Many of us would have been more comfortable with a more classic design that altered only one variable, either the pressure or the wound interface. Despite the limitations, this project offers proof of concept and insights into the effectiveness of a novel negative-pressure wound therapy approach with low pressure and a wound dressing that is not the traditional foam dressing. Many of us expected the lower pressure and new dressing to fail. Our wound team cringed at the thought of not being able to use the higher pressure levels they had been taught to use to heal wounds from their earliest experiences with negative-pressure wound therapy. Thus, in several respects, the approach with low pressure and a silicone gauze interface contradicts our previous belief system.

Although negative-pressure wound therapy has significantly improved the treatment and outcomes of complex wounds, clinicians still have little insight into how to dose pressure or what wound interfaces to use to optimize therapy. Additional large randomized clinical studies that compare systems currently in clinical use or studies that compare pressures or interfaces are needed to aid clinicians in choosing the best options for patient care.

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A: A good rule of thumb is as follows (these papers are not amenable to LOE grading):

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- Instructional course lectures
- CME courses
- Editorials
- Correspondence

As far as what is or is not ratable, the standard is to exclude basic science, bench work, animal, and cadaveric studies because the information gained from these studies is not something that can be applied directly to patient treatment decisions.

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